

UNAM – Computer Programming I (I3691CP)

Group Pitch Summary – Semester Project

Group Composition

Total Members: 14

Roles Assigned

Project Manager: **Pombili_Hamwoomo_225067250**

Lead Developer: **Matheus_T_Medusalem_224061321**

Firebase Lead: **Ndahambelela_Petrus_225145693**

UI/UX Lead: **John_M_HAIKALI_224030523**

Documentation Lead: **Amadhila_S.M_225119919**

GitHub Manager: **Nakashimba_Lahya_225089831**

Assistant Developers (1): **Max_Simon_225009145**

Assistant Developers (2): **Dortia_Ashaama_222124253**

Firebase Assistants (1): **Josephat_Amoomo_225086972**

Firebase Assistants (2): **Stephanson_Tuukondjele_225077264**

UI/UX Designers (1): **Ruben_Paulus_222167335**

UI/UX Designers (2): **Mukuwe_Mathias_225070553**

Testing & QA Lead: **Hamunyela_Penny P 225058219**

Presentation & Communication Lead: **Tuyeni_Helvi_225067080**

Idea 1: Engineering Task Management & Dashboard App

Problem: Engineering officers face time mismanagement, data loss, and difficulty tracking multiple tasks manually.

Target Users: Engineering officers and supervisors.

Core Features:

- Task dashboard showing all tasks with priorities.
- Notifications/reminders via Firebase Cloud Messaging.
- Secure task storage in Firebase Firestore.
- Authentication with Firebase for role-based access.
- Reports of completed tasks for accountability.

Impact: Reduces time mismanagement, prevents data loss, improves productivity, and enhances accountability.

Idea 2: Shift Handover Digital Log

Problem: Poor documentation during shift handovers causes miscommunication and repeated work.

Target Users: Outgoing and incoming shift workers, supervisors.

Core Features:

- Workers log shift details (tasks, equipment status, issues).
- Reports stored in Firestore, accessible to the next shift.
- Authentication ensures secure access.
- Optional photo attachments stored in Firebase Storage.

Impact: Smooth communication, better coordination, improved productivity.

Idea 3: Underground Hazard & Emergency Communication App

Problem: Emergency response underground is slow due to poor communication.

Target Users: Miners and supervisors.

Core Features:

- SOS emergency alert system.
- Hazard reporting stored in Firestore.
- Location tagging for incidents.
- Push notifications via Firebase Cloud Messaging.

Impact: Faster emergency response, improved safety awareness, better hazard tracking.